ECROS-5400UV: LCD Screen Specification

Version: 1.0

Date: 2026-04-15

> **Source of truth:** This document is generated from src/domain/constants/screens.json. Source Screen IDs from txt files are preserved as traceability metadata and are not used as unique keys.

Unified Indexing Rules

Unified index	runtime id	Legacy WND	Title	Mode	Kind
SYS-BOOT	boot	WND-01	Welcome Window	System	System
SYS-DIAGNOSTIC	diagnostic	WND-02	Diagnostics	System	System
SYS-WARMUP	warmup	WND-03	Warm-Up	System	System

Unified index	runtime id	Legacy WND	Title	Mode	Kind
PHOT-MAIN	photometry	WND-05	Photometry	Photometry	Measurement
PHOT-SIGNAL	photometryValue	WND-06	Photometry Signal Selection	Photometry	Menu
PHOT-GRAPH	photometryGraph	WND-07	Photometry Graph	Photometry	Graph
QUANT-MAIN	quantMain	WND-13	Quantitative Analysis	Quantitative analysis	Menu
QUANT-UNITS	quantUnits	WND-14	Measurement Units	Quantitative analysis	Menu
QUANT-COEF	quantCoef	WND-15	Coefficients	Quantitative analysis	Measurement
QUANT-CAL-STANDARDS	calibrationSetupStandards	WND-16	New Calibration: Standards	Quantitative analysis	Input
QUANT-CAL-PARALLELS	calibrationSetupParallels	WND-17	New Calibration: Replicates	Quantitative analysis	Input
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QUANT-CAL-STEP	calibrationStep	WND-19	Calibration Step	Quantitative analysis	Measurement
QUANT-CAL-JOURNAL	calibrationJournal	WND-20	Calibration Journal	Quantitative analysis	Measurement
QUANT-CAL-GRAPH	calibrationGraph	WND-21	Calibration Graph	Quantitative analysis	Graph
KIN-MENU	kineticsMenu	WND-22	Kinetics Parameters	Kinetics	Menu
KIN-RUN	kineticsRun	WND-23	Kinetics Measurement	Kinetics	Measurement
KIN-GRAPH	kineticsGraph	WND-24	Kinetics Graph	Kinetics	Graph
MW-MENU	multiWaveMenu	WND-25	Multi-Wavelength Analysis	Multi-wavelength analysis	Menu
MW-SETUP	multiWaveSetup	WND-26	Wavelength Setup	Multi-wavelength analysis	Input
MW-RUN	multiWaveRun	WND-27	Multi-Wavelength Measurement	Multi-wavelength analysis	Measurement
MW-JOURNAL	multiWaveJournal	WND-28	Multi-Wavelength Journal	Multi-wavelength analysis	Measurement
MW-GRAPH	multiWaveGraph	WND-29	Multi-Wavelength Graph	Multi-wavelength analysis	Graph

Unified index	runtime id	Legacy WND	Title	Mode	Kind
SET-MAIN	settings	WND-30	Settings	Settings	Menu
SET-VERSION	version	WND-31	Instrument Version	Settings	System
FILE-ROOT	fileRoot	WND-08	File Groups	Files	File
FILE-LIST	fileList	WND-09	File List	Files	File
FILE-ACTION	fileActionMenu	WND-10	File Actions	Files	File
FILE-SAVE	saveDialog	WND-11	Save File	Files	Dialog
MAIN-MENU	main	WND-04	Main Menu	Shared screens	Menu
SHARED-INPUT	input	WND-12	Data Input	Shared screens	Input
SHARED-WARNING	warning	WND-32	Warning	Shared screens	Dialog

Global State Variables

Field	Source/usage
`state.screen`	runtime id of the active window
`inputTarget`	current input target
`returnScreen`	return screen after a dialog
`warningReturn`	return screen after a warning
`previousScreen`	previous screen in the route
`saveMeta.group`	saved file group
`fileContext.group`	active file-manager group

Screens

SYS-BOOT · Welcome Window

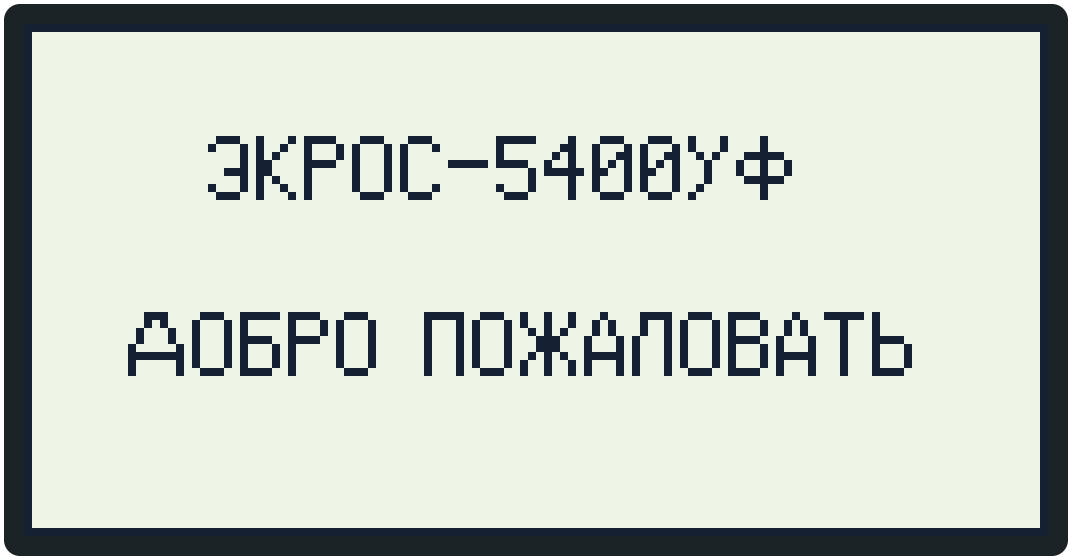
ID
boot

LEGACY WND
WND-01

MODE
System

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Welcome Window is used as a system in the "System" group.

Information meaning: The window captures the "System" scenario state and links runtime id "boot" to unified index SYS-BOOT.

Content: LCD content is assembled from fields: logo, softwareVersion.

Relation to other windows: Outgoing routes lead to: diagnostic.

Source Screen IDs: 1-1

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`logo`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "boot".
`softwareVersion`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "boot".

Transition routes

Action/button	Guard	Operation	Target screen
timer	—	runtime boot timer	diagnostic · Diagnostics

SYS-DIAGNOSTIC · Diagnostics

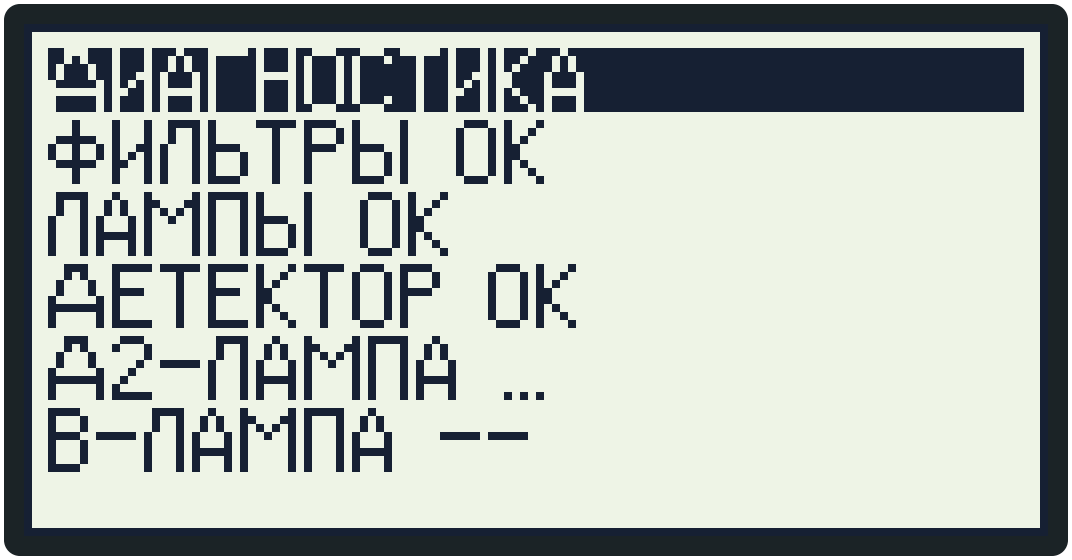
ID
diagnostic

LEGACY WND
WND-02

MODE
System

STATUS
implemented

LCD mockup

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[object Object]
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[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Diagnostics is used as a system in the "System" group.

Information meaning: The window captures the "System" scenario state and links runtime id "diagnostic" to unified index SYS-DIAGNOSTIC.

Content: LCD content is assembled from fields: filterStatus, lampStatus, detectorStatus, d2LampStatus, wLampStatus, wavelengthCalibrationStatus, darkCurrentStatus.

Relation to other windows: Outgoing routes lead to: warmup.

Source Screen IDs: 1-2

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`filterStatus`	string node or check status	Used by LCD renderer or handlers for screen "diagnostic".
`lampStatus`	string node or check status	Used by LCD renderer or handlers for screen "diagnostic".
`detectorStatus`	string node or check status	Used by LCD renderer or handlers for screen "diagnostic".
`d2LampStatus`	string node or check status	Used by LCD renderer or handlers for screen "diagnostic".
`wLampStatus`	string node or check status	Used by LCD renderer or handlers for screen "diagnostic".
`wavelengthCalibrationStatus`	string node or check status	Used by LCD renderer or handlers for screen "diagnostic".
`darkCurrentStatus`	string node or check status	Used by LCD renderer or handlers for screen "diagnostic".

Transition routes

Action/button	Guard	Operation	Target screen
done	—	runtime diagnostic complete	warmup · Warm-Up

SYS-WARMUP · Warm-Up

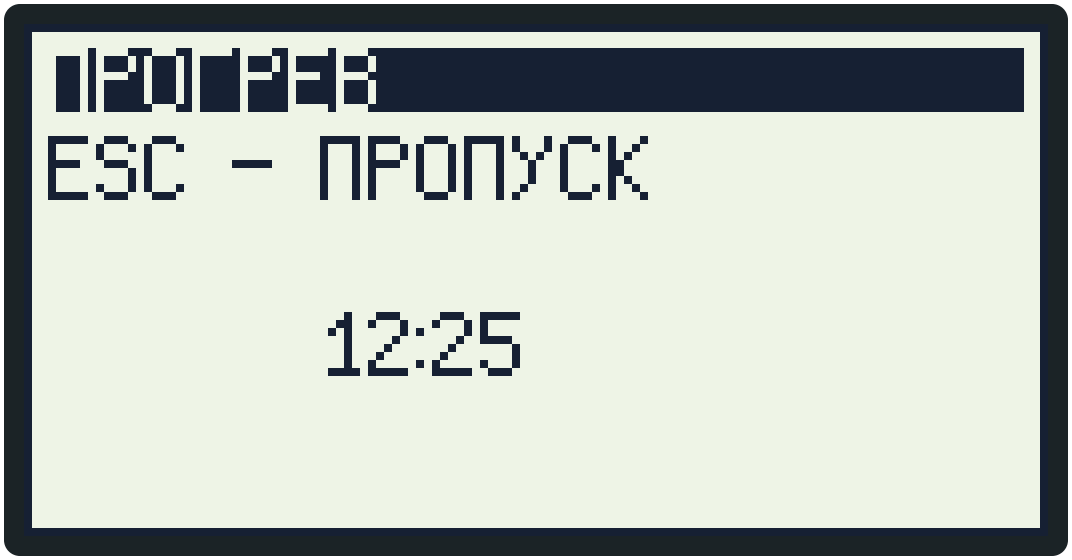
ID
warmup

LEGACY WND
WND-03

MODE
System

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Warm-Up is used as a system in the "System" group.

Information meaning: The window captures the "System" scenario state and links runtime id "warmup" to unified index SYS-WARMUP.

Content: LCD content is assembled from fields: warmupRemaining, skipHint.

Relation to other windows: Outgoing routes lead to: main.

Source Screen IDs: 1-3

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`warmupRemaining`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "warmup".
`skipHint`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "warmup".

Transition routes

Action/button	Guard	Operation	Target screen
esc/timer	—	runtime warmup	main · Main Menu

PHOT-MAIN · Photometry

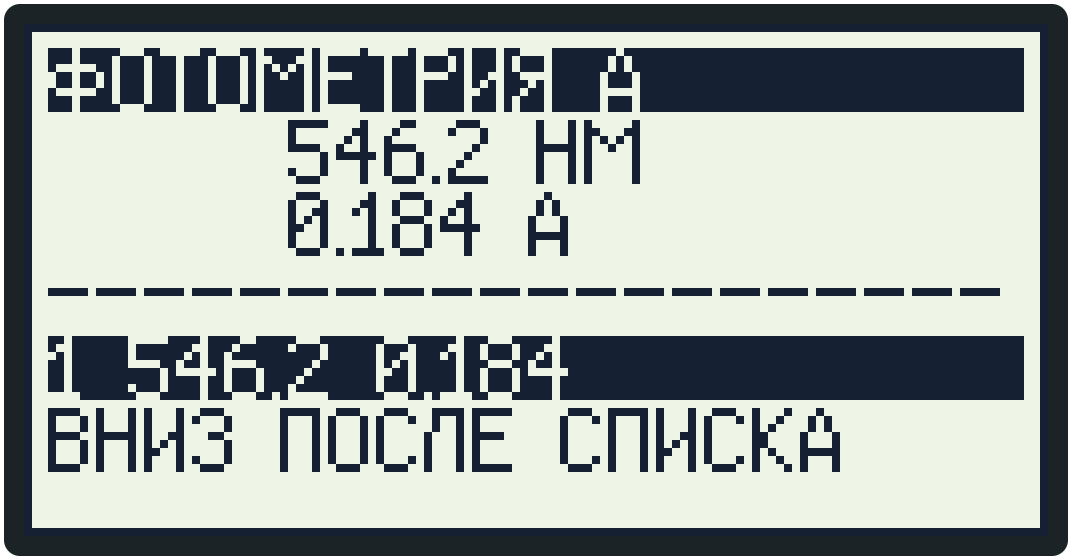
ID
photometry

LEGACY WND
WND-05

MODE
Photometry

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Photometry is used as a measurement in the "Photometry" group.

Information meaning: The window captures the "Photometry" scenario state and links runtime id "photometry" to unified index PHOT-MAIN.

Content: LCD content is assembled from fields: wavelength, selectedSignal, lastComputedA, lastComputedT, lastEnergy, measurements.

Relation to other windows: Outgoing routes lead to: photometryValue, photometryGraph, input, fileList, saveDialog, main.

Source Screen IDs: 1-5-1 1-5-2 1-5-3

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`wavelength`	number, wavelength in nm	Used by LCD renderer or handlers for screen "photometry".
`selectedSignal`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "photometry".
`lastComputedA`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "photometry".
`lastComputedT`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "photometry".
`lastEnergy`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "photometry".
`measurements`	measurement result structure	Used by LCD renderer or handlers for screen "photometry".

Transition routes

Action/button	Guard	Operation	Target screen
set	—	value type selection	photometryValue · Photometry Signal Selection
down	measurementCursor at last row	runtime graph shortcut	photometryGraph · Photometry Graph
goto λ	—	Screen 1-5-1 task установить λ	input · Data Input
file	no measurements	runtime browse files	fileList · File List
file	measurements exist	runtime save measurements	saveDialog · Save File
esc	—	runtime escape	main · Main Menu

PHOT-SIGNAL · Photometry Signal Selection

ID
photometryValue

LEGACY WND
WND-06

MODE
Photometry

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Photometry Signal Selection is used as a menu in the "Photometry" group.

Information meaning: The window captures the "Photometry" scenario state and links runtime id "photometryValue" to unified index PHOT-SIGNAL.

Content: LCD content is assembled from fields: signalOptions, photometryValueIndex.

Relation to other windows: Outgoing routes lead to: photometry, multiWaveMenu.

Source Screen IDs: 1-5-1-2 1-5-1-2-1 1-5-1-2-4

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`signalOptions`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "photometryValue".
`photometryValueIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "photometryValue".

Transition routes

Action/button	Guard	Operation	Target screen
enter/esc	returnScreen == photometry	runtime return	photometry · Photometry
enter/esc	returnScreen == multiWaveMenu	runtime return from multiwave	multiWaveMenu · Multi-Wavelength Analysis

PHOT-GRAPH · Photometry Graph

ID
photometryGraph

LEGACY WND
WND-07

MODE
Photometry

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Photometry Graph is used as a graph in the "Photometry" group.

Information meaning: The window captures the "Photometry" scenario state and links runtime id "photometryGraph" to unified index PHOT-GRAPH.

Content: LCD content is assembled from fields: measurementsGraph, wavelength, lastComputedA.

Relation to other windows: Outgoing routes lead to: photometry, saveDialog.

Source Screen IDs: 1-5-1-8 1-5-1-9

Source integrity: reindexed

State table

Field	Type/meaning	Source/usage
`measurementsGraph`	array of graph points	Used by LCD renderer or handlers for screen "photometryGraph".
`wavelength`	number, wavelength in nm	Used by LCD renderer or handlers for screen "photometryGraph".
`lastComputedA`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "photometryGraph".

Transition routes

Action/button	Guard	Operation	Target screen
esc	—	runtime return to journal	photometry · Photometry
file	—	runtime save graph	saveDialog · Save File

QUANT-MAIN · Quantitative Analysis

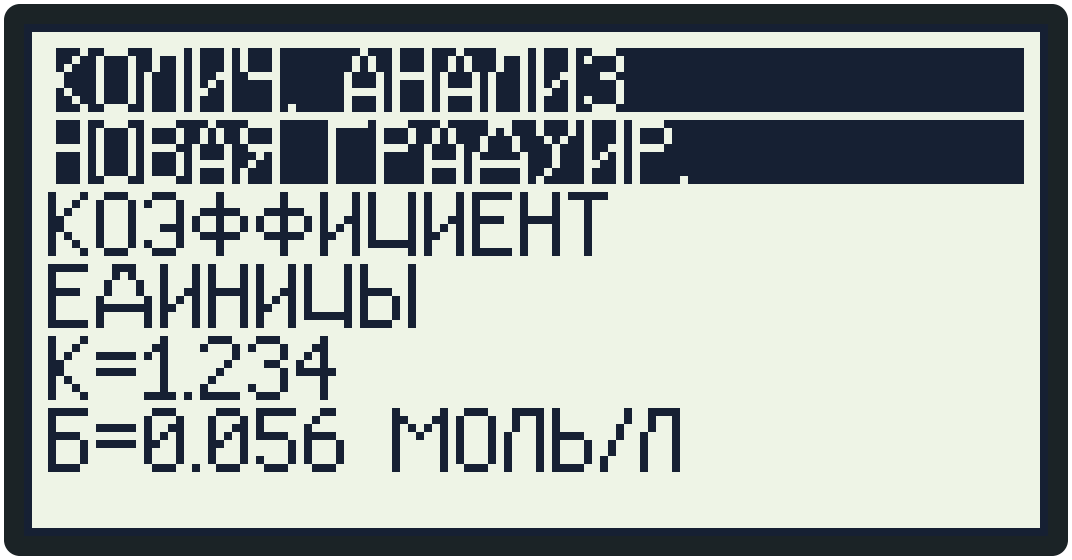
ID
quantMain

LEGACY WND
WND-13

MODE
Quantitative analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Quantitative Analysis is used as a menu in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "quantMain" to unified index QUANT-MAIN.

Content: LCD content is assembled from fields: quantMenu, quantIndex, quantK, quantB, unitsIndex.

Relation to other windows: Outgoing routes lead to: calibrationSetupStandards, quantCoef, quantUnits, fileList, main.

Source Screen IDs: 1-6-1 1-6-2

Source integrity: ambiguous

State table

Field	Type/meaning	Source/usage
`quantMenu`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantMain".
`quantIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "quantMain".
`quantK`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantMain".
`quantB`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantMain".
`unitsIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "quantMain".

Transition routes

Action/button	Guard	Operation	Target screen
enter	quantIndex == 0	new calibration	calibrationSetupStandards · New Calibration: Standards
enter	quantIndex == 1	coefficient method	quantCoef · Coefficients
enter	quantIndex == 2	units selection	quantUnits · Measurement Units
file	—	runtime browse calibration files	fileList · File List
esc	—	runtime escape	main · Main Menu

QUANT-UNITS · Measurement Units

ID
quantUnits

LEGACY WND
WND-14

MODE
Quantitative analysis

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Measurement Units is used as a menu in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "quantUnits" to unified index QUANT-UNITS.

Content: LCD content is assembled from fields: units, unitsIndex, returnScreen.

Relation to other windows: Outgoing routes lead to: quantMain, multiWaveMenu.

Source Screen IDs: 1-6-1

Source integrity: ambiguous

State table

Field	Type/meaning	Source/usage
`units`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantUnits".
`unitsIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "quantUnits".
`returnScreen`	contextual return screen	Used by LCD renderer or handlers for screen "quantUnits".

Transition routes

Action/button	Guard	Operation	Target screen
enter/esc	returnScreen == quantMain	runtime return	quantMain · Quantitative Analysis
enter/esc	returnScreen == multiWaveMenu	runtime return from multiwave	multiWaveMenu · Multi-Wavelength Analysis

QUANT-COEF · Coefficients

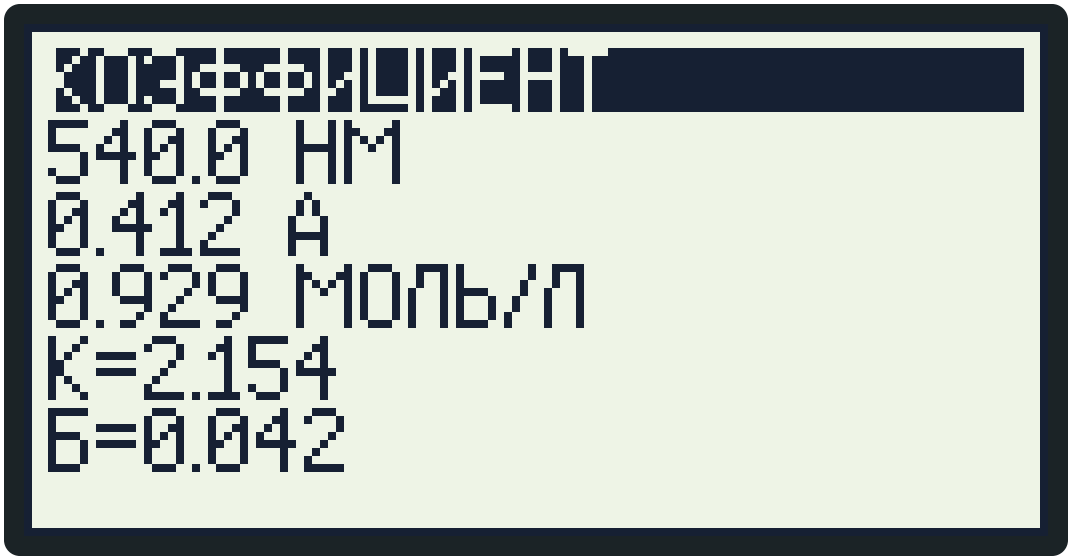
ID
quantCoef

LEGACY WND
WND-15

MODE
Quantitative analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Coefficients is used as a measurement in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "quantCoef" to unified index QUANT-COEF.

Content: LCD content is assembled from fields: wavelength, lastComputedA, concentration, quantK, quantB, unitsIndex.

Relation to other windows: Outgoing routes lead to: input, saveDialog, quantMain.

Source Screen IDs: 1-6-1 1-6-1-1-32 1-6-1-1-33 1-6-1-1-34

Source integrity: ambiguous

State table

Field	Type/meaning	Source/usage
`wavelength`	number, wavelength in nm	Used by LCD renderer or handlers for screen "quantCoef".
`lastComputedA`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantCoef".
`concentration`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantCoef".
`quantK`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantCoef".
`quantB`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "quantCoef".
`unitsIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "quantCoef".

Transition routes

Action/button	Guard	Operation	Target screen
1/2/goto λ	—	coefficient or wavelength input	input · Data Input
file	—	runtime save coefficient	saveDialog · Save File
esc	—	runtime return	quantMain · Quantitative Analysis

QUANT-CAL-STANDARDS · New Calibration: Standards

ID
calibrationSetupStandards

LEGACY WND
WND-16

MODE
Quantitative analysis

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: New Calibration: Standards is used as a input in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "calibrationSetupStandards" to unified index QUANT-CAL-STANDARDS.

Content: LCD content is assembled from fields: calibration.standards.

Relation to other windows: Outgoing routes lead to: calibrationSetupParallels, quantMain.

Source Screen IDs: 1-6-1-1 1-6-1-1-2

Source integrity: reindexed

State table

Field	Type/meaning	Source/usage
`calibration.standards`	calibration state	Used by LCD renderer or handlers for screen "calibrationSetupStandards".

Transition routes

Action/button	Guard	Operation	Target screen
enter	—	runtime next setup step	calibrationSetupParallels · New Calibration: Replicates
esc	—	runtime return	quantMain · Quantitative Analysis

QUANT-CAL-PARALLELS · New Calibration: Replicates

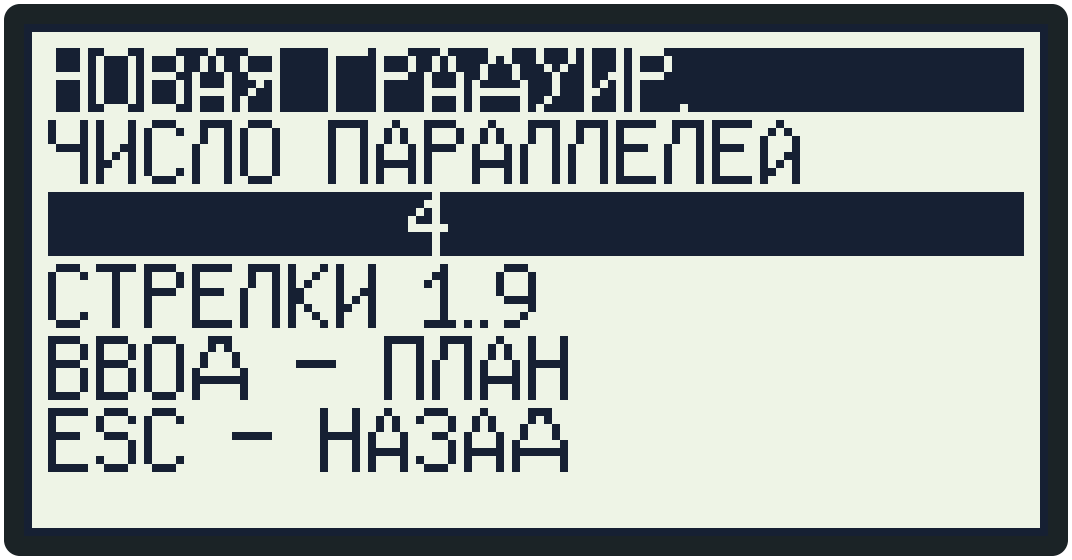
ID
calibrationSetupParallels

LEGACY WND
WND-17

MODE
Quantitative analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: New Calibration: Replicates is used as a input in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "calibrationSetupParallels" to unified index QUANT-CAL-PARALLELS.

Content: LCD content is assembled from fields: calibration.parallels.

Relation to other windows: Outgoing routes lead to: calibrationPlan, calibrationSetupStandards.

Source Screen IDs: 1-6-1-1-3 1-6-1-1-4

Source integrity: reindexed

State table

Field	Type/meaning	Source/usage
`calibration.parallels`	calibration state	Used by LCD renderer or handlers for screen "calibrationSetupParallels".

Transition routes

Action/button	Guard	Operation	Target screen
enter	—	runtime build calibration plan	calibrationPlan · Calibration Measurement Plan
esc	—	runtime previous setup step	calibrationSetupStandards · New Calibration: Standards

QUANT-CAL-PLAN · Calibration Measurement Plan

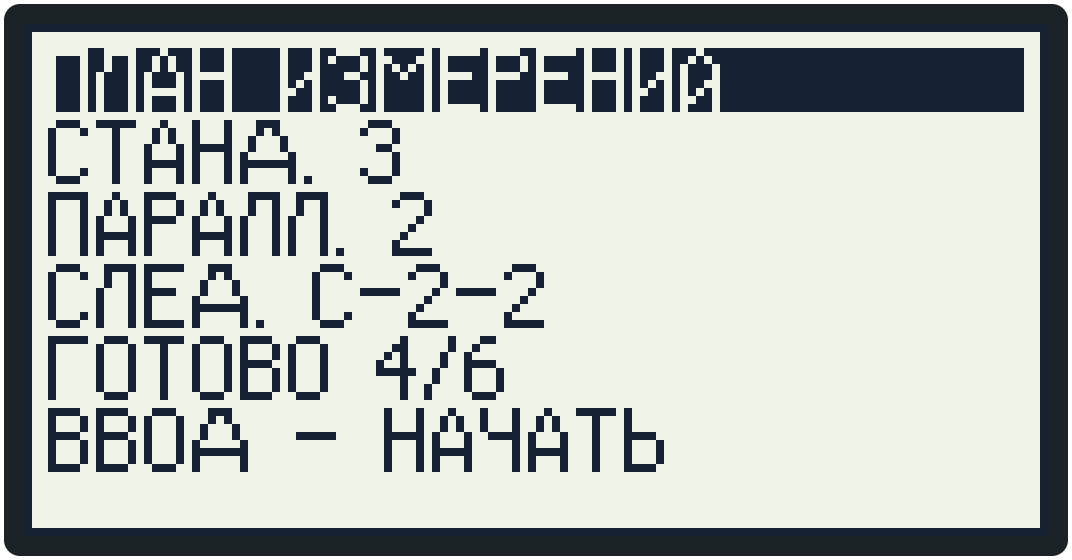
ID
calibrationPlan

LEGACY WND
WND-18

MODE
Quantitative analysis

STATUS
implemented

LCD mockup

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[object Object]
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[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Calibration Measurement Plan is used as a menu in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "calibrationPlan" to unified index QUANT-CAL-PLAN.

Content: LCD content is assembled from fields: calibration.plan, calibration.stepIndex, calibration done count.

Relation to other windows: Outgoing routes lead to: calibrationStep, calibrationJournal, fileList, quantMain.

Source Screen IDs: 1-6-1-1-5 1-6-1-1-6

Source integrity: reindexed

State table

Field	Type/meaning	Source/usage
`calibration.plan`	calibration state	Used by LCD renderer or handlers for screen "calibrationPlan".
`calibration.stepIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "calibrationPlan".
`calibration done count`	calibration state	Used by LCD renderer or handlers for screen "calibrationPlan".

Transition routes

Action/button	Guard	Operation	Target screen
enter	—	runtime start measuring plan	calibrationStep · Calibration Step
down	measured results exist	runtime open journal	calibrationJournal · Calibration Journal
file	—	runtime browse calibration files	fileList · File List
esc	—	runtime return	quantMain · Quantitative Analysis

QUANT-CAL-STEP · Calibration Step

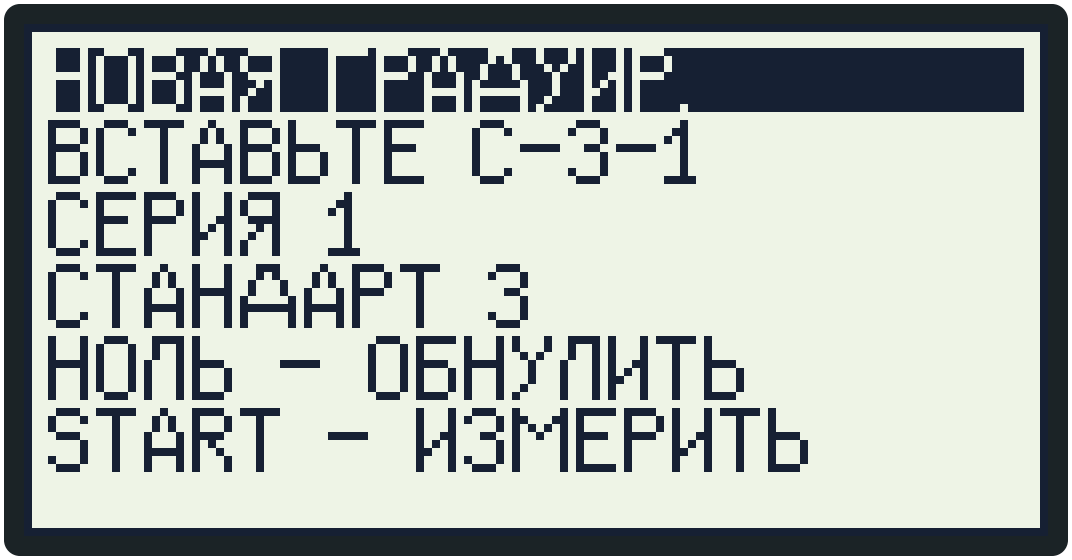
ID
calibrationStep

LEGACY WND
WND-19

MODE
Quantitative analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Calibration Step is used as a measurement in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "calibrationStep" to unified index QUANT-CAL-STEP.

Content: LCD content is assembled from fields: calibration current standard, calibration current parallel, measurement actions.

Relation to other windows: Outgoing routes lead to: calibrationPlan, calibrationJournal, saveDialog.

Source Screen IDs: 1-6-1-1-7 1-6-1-1-8 1-6-1-1-9 1-6-1-1-10 1-6-1-1-11 1-6-1-1-12 1-6-1-1-13 1-6-1-1-14 1-6-1-1-15 1-6-1-1-16 1-6-1-1-17 1-6-1-1-18 1-6-1-1-19 1-6-1-1-20 1-6-1-1-21 1-6-1-1-22 1-6-1-1-23 1-6-1-1-24 1-6-1-1-25 1-6-1-1-26

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`calibration current standard`	calibration state	Used by LCD renderer or handlers for screen "calibrationStep".
`calibration current parallel`	calibration state	Used by LCD renderer or handlers for screen "calibrationStep".
`measurement actions`	measurement result structure	Used by LCD renderer or handlers for screen "calibrationStep".

Transition routes

Action/button	Guard	Operation	Target screen
esc	—	runtime return to plan	calibrationPlan · Calibration Measurement Plan
measure	—	runtime measured calibration step	calibrationJournal · Calibration Journal
file	—	runtime save calibration	saveDialog · Save File

QUANT-CAL-JOURNAL · Calibration Journal

ID
calibrationJournal

LEGACY WND
WND-20

MODE
Quantitative analysis

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Calibration Journal is used as a measurement in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "calibrationJournal" to unified index QUANT-CAL-JOURNAL.

Content: LCD content is assembled from fields: calibration results, calibration.resultCursor.

Relation to other windows: Outgoing routes lead to: calibrationPlan, calibrationStep, calibrationGraph, saveDialog.

Source Screen IDs: 1-6-1-1-27 1-6-1-1-28 1-6-1-1-29 1-6-1-1-30 1-6-1-1-31

Source integrity: reindexed

State table

Field	Type/meaning	Source/usage
`calibration results`	calibration state	Used by LCD renderer or handlers for screen "calibrationJournal".
`calibration.resultCursor`	calibration state	Used by LCD renderer or handlers for screen "calibrationJournal".

Transition routes

Action/button	Guard	Operation	Target screen
esc	—	runtime return to plan	calibrationPlan · Calibration Measurement Plan
enter/start/clear	—	runtime continue or remeasure	calibrationStep · Calibration Step
down/end	—	runtime graph after journal	calibrationGraph · Calibration Graph
file	—	runtime save calibration journal	saveDialog · Save File

QUANT-CAL-GRAPH · Calibration Graph

ID
calibrationGraph

LEGACY WND
WND-21

MODE
Quantitative analysis

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Calibration Graph is used as a graph in the "Quantitative analysis" group.

Information meaning: The window captures the "Quantitative analysis" scenario state and links runtime id "calibrationGraph" to unified index QUANT-CAL-GRAPH.

Content: LCD content is assembled from fields: calibration graph values.

Relation to other windows: Outgoing routes lead to: calibrationJournal, saveDialog.

Source Screen IDs: 1-6-1-1-32 1-6-1-1-33 1-6-1-1-34

Source integrity: reindexed

State table

Field	Type/meaning	Source/usage
`calibration graph values`	array of graph points	Used by LCD renderer or handlers for screen "calibrationGraph".

Transition routes

Action/button	Guard	Operation	Target screen
esc	—	runtime return to journal	calibrationJournal · Calibration Journal
file	—	runtime save calibration graph	saveDialog · Save File

KIN-MENU · Kinetics Parameters

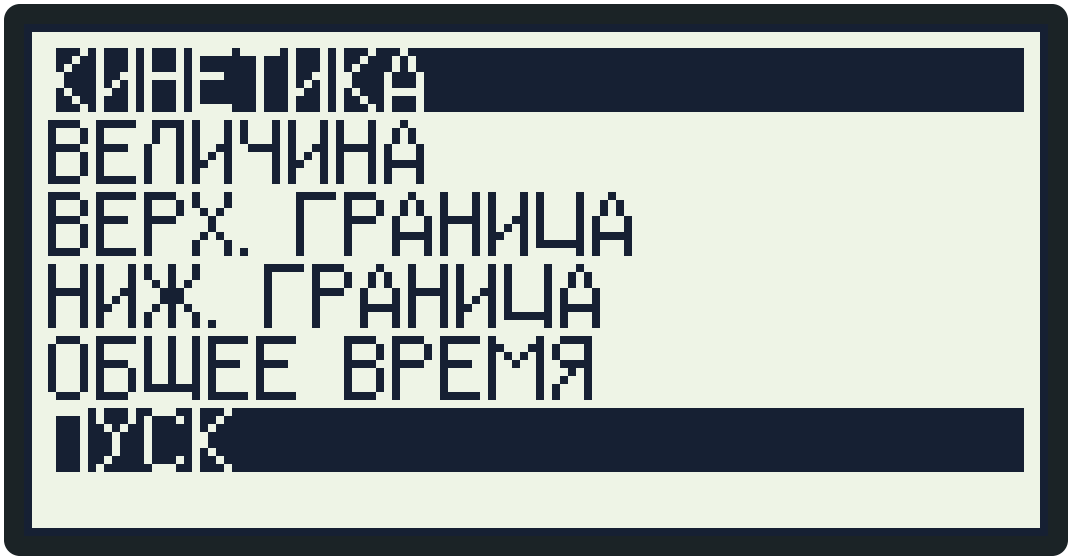
ID
kineticsMenu

LEGACY WND
WND-22

MODE
Kinetics

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Kinetics Parameters is used as a menu in the "Kinetics" group.

Information meaning: The window captures the "Kinetics" scenario state and links runtime id "kineticsMenu" to unified index KIN-MENU.

Content: LCD content is assembled from fields: kinetics menu, kineticsIndex, kineticDuration, kineticUpper, kineticLower.

Relation to other windows: Outgoing routes lead to: input, kineticsRun, fileList, main.

Source Screen IDs: 1-4-1 1-4-1-1

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`kinetics menu`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsMenu".
`kineticsIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "kineticsMenu".
`kineticDuration`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsMenu".
`kineticUpper`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsMenu".
`kineticLower`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsMenu".

Transition routes

Action/button	Guard	Operation	Target screen
enter upper/lower/duration	—	kinetics parameter input	input · Data Input
start	—	runtime start kinetics	kineticsRun · Kinetics Measurement
file	—	runtime browse kinetics files	fileList · File List
esc	—	runtime return	main · Main Menu

KIN-RUN · Kinetics Measurement

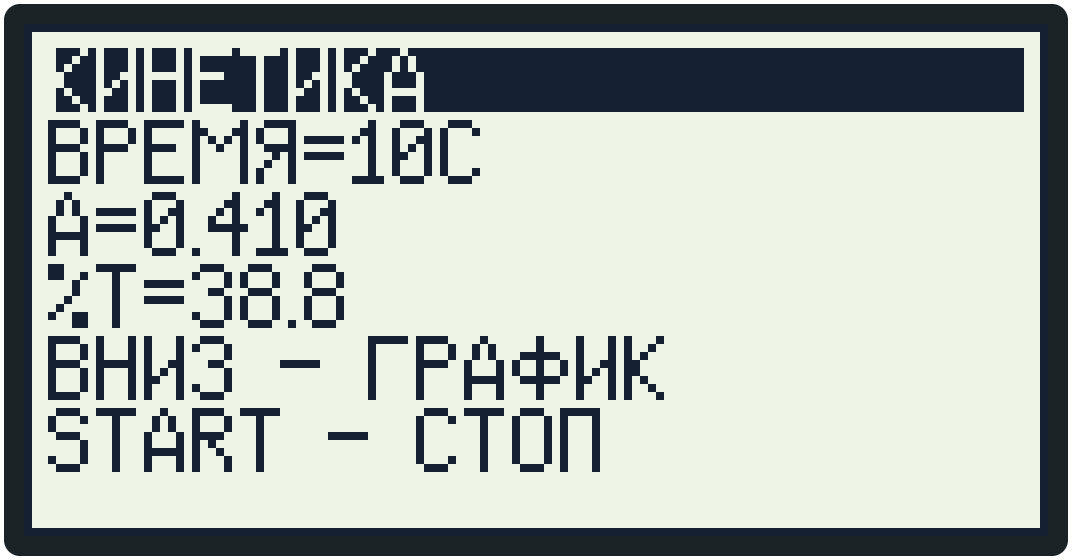
ID
kineticsRun

LEGACY WND
WND-23

MODE
Kinetics

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Kinetics Measurement is used as a measurement in the "Kinetics" group.

Information meaning: The window captures the "Kinetics" scenario state and links runtime id "kineticsRun" to unified index KIN-RUN.

Content: LCD content is assembled from fields: kineticPoints, lastComputedA, lastComputedT, kinetic time.

Relation to other windows: Outgoing routes lead to: kineticsGraph, kineticsMenu, saveDialog.

Source Screen IDs: 1-6-1-1-27

Source integrity: ambiguous

State table

Field	Type/meaning	Source/usage
`kineticPoints`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsRun".
`lastComputedA`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsRun".
`lastComputedT`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsRun".
`kinetic time`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsRun".

Transition routes

Action/button	Guard	Operation	Target screen
down	—	runtime open graph	kineticsGraph · Kinetics Graph
stop/esc	—	runtime stop kinetics	kineticsMenu · Kinetics Parameters
file	—	runtime save kinetics	saveDialog · Save File

KIN-GRAPH · Kinetics Graph

ID
kineticsGraph

LEGACY WND
WND-24

MODE
Kinetics

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Kinetics Graph is used as a graph in the "Kinetics" group.

Information meaning: The window captures the "Kinetics" scenario state and links runtime id "kineticsGraph" to unified index KIN-GRAPH.

Content: LCD content is assembled from fields: kinetic graph, kineticUpper, kineticLower.

Relation to other windows: Outgoing routes lead to: kineticsRun, saveDialog.

Source Screen IDs: 1-6-1-1-27

Source integrity: ambiguous

State table

Field	Type/meaning	Source/usage
`kinetic graph`	array of graph points	Used by LCD renderer or handlers for screen "kineticsGraph".
`kineticUpper`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsGraph".
`kineticLower`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "kineticsGraph".

Transition routes

Action/button	Guard	Operation	Target screen
esc	—	runtime return to run	kineticsRun · Kinetics Measurement
file	—	runtime save kinetics graph	saveDialog · Save File

MW-MENU · Multi-Wavelength Analysis

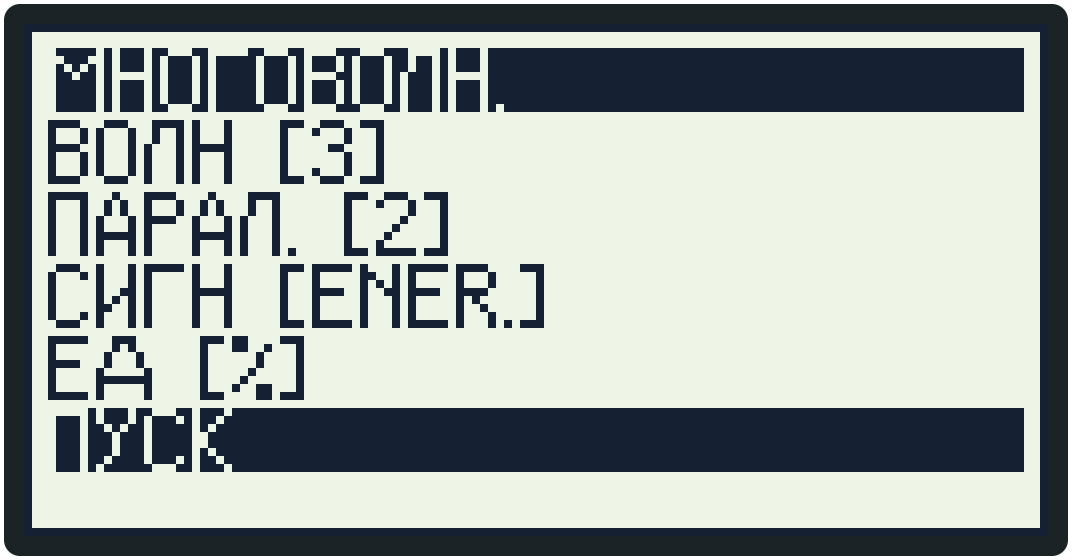
ID
multiWaveMenu

LEGACY WND
WND-25

MODE
Multi-wavelength analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Multi-Wavelength Analysis is used as a menu in the "Multi-wavelength analysis" group.

Information meaning: The window captures the "Multi-wavelength analysis" scenario state and links runtime id "multiWaveMenu" to unified index MW-MENU.

Content: LCD content is assembled from fields: multiWaveCount, multiWaveParallelCount, photometryValueIndex, unitsIndex, multiWaveIndex.

Relation to other windows: Outgoing routes lead to: input, multiWaveSetup, photometryValue, quantUnits, multiWaveRun, fileList, main.

Source Screen IDs: multi-wave-menu

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`multiWaveCount`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "multiWaveMenu".
`multiWaveParallelCount`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "multiWaveMenu".
`photometryValueIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "multiWaveMenu".
`unitsIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "multiWaveMenu".
`multiWaveIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "multiWaveMenu".

Transition routes

Action/button	Guard	Operation	Target screen
count/parallel/goto λ	—	runtime multiwave input	input · Data Input
enter count	—	runtime configure wavelengths	multiWaveSetup · Wavelength Setup
signal	—	runtime choose signal	photometryValue · Photometry Signal Selection
units	—	runtime choose units	quantUnits · Measurement Units
start	—	runtime start multiwave	multiWaveRun · Multi-Wavelength Measurement
file	—	runtime browse multiwave files	fileList · File List
esc	—	runtime return	main · Main Menu

MW-SETUP · Wavelength Setup

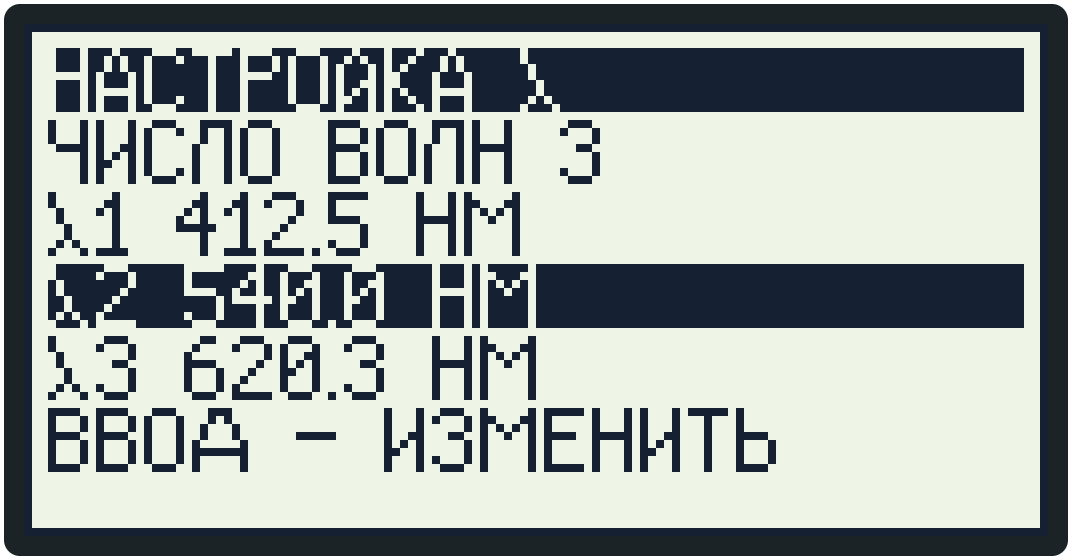
ID
multiWaveSetup

LEGACY WND
WND-26

MODE
Multi-wavelength analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Wavelength Setup is used as a input in the "Multi-wavelength analysis" group.

Information meaning: The window captures the "Multi-wavelength analysis" scenario state and links runtime id "multiWaveSetup" to unified index MW-SETUP.

Content: LCD content is assembled from fields: multiWaveWavelengths, multiWaveSetupIndex.

Relation to other windows: Outgoing routes lead to: input, multiWaveMenu.

Source Screen IDs: multi-wave-setup

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`multiWaveWavelengths`	number, wavelength in nm	Used by LCD renderer or handlers for screen "multiWaveSetup".
`multiWaveSetupIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "multiWaveSetup".

Transition routes

Action/button	Guard	Operation	Target screen
edit λN	—	runtime edit wavelength	input · Data Input
esc	—	runtime return	multiWaveMenu · Multi-Wavelength Analysis

MW-RUN · Multi-Wavelength Measurement

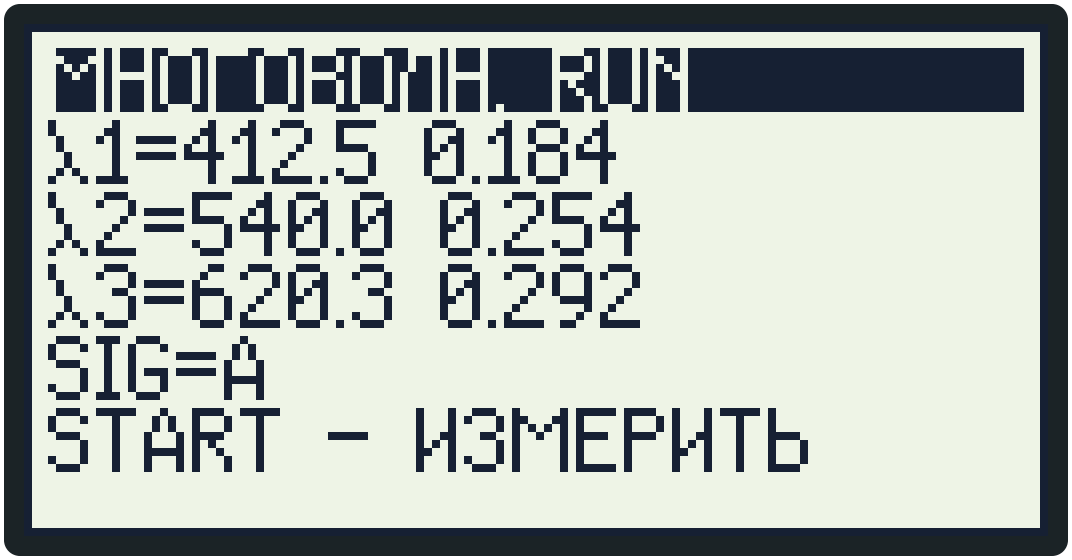
ID
multiWaveRun

LEGACY WND
WND-27

MODE
Multi-wavelength analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Multi-Wavelength Measurement is used as a measurement in the "Multi-wavelength analysis" group.

Information meaning: The window captures the "Multi-wavelength analysis" scenario state and links runtime id "multiWaveRun" to unified index MW-RUN.

Content: LCD content is assembled from fields: multiWaveGraphData, multiWaveMeasurements, selectedSignal.

Relation to other windows: Outgoing routes lead to: multiWaveJournal, saveDialog, fileList, multiWaveMenu.

Source Screen IDs: multi-wave-run

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`multiWaveGraphData`	array of graph points	Used by LCD renderer or handlers for screen "multiWaveRun".
`multiWaveMeasurements`	measurement result structure	Used by LCD renderer or handlers for screen "multiWaveRun".
`selectedSignal`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "multiWaveRun".

Transition routes

Action/button	Guard	Operation	Target screen
measure	—	runtime measured run	multiWaveJournal · Multi-Wavelength Journal
file save	measurements exist	runtime save multiwave	saveDialog · Save File
file browse	no measurements	runtime browse multiwave	fileList · File List
esc	—	runtime return	multiWaveMenu · Multi-Wavelength Analysis

MW-JOURNAL · Multi-Wavelength Journal

ID
multiWaveJournal

LEGACY WND
WND-28

MODE
Multi-wavelength analysis

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Multi-Wavelength Journal is used as a measurement in the "Multi-wavelength analysis" group.

Information meaning: The window captures the "Multi-wavelength analysis" scenario state and links runtime id "multiWaveJournal" to unified index MW-JOURNAL.

Content: LCD content is assembled from fields: multiWaveMeasurements, multiWaveMeasurementCursor.

Relation to other windows: Outgoing routes lead to: multiWaveRun, multiWaveGraph, saveDialog.

Source Screen IDs: multi-wave-journal

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`multiWaveMeasurements`	measurement result structure	Used by LCD renderer or handlers for screen "multiWaveJournal".
`multiWaveMeasurementCursor`	measurement result structure	Used by LCD renderer or handlers for screen "multiWaveJournal".

Transition routes

Action/button	Guard	Operation	Target screen
enter/esc	—	runtime return to run	multiWaveRun · Multi-Wavelength Measurement
down	—	runtime open graph	multiWaveGraph · Multi-Wavelength Graph
file	—	runtime save journal	saveDialog · Save File

MW-GRAPH · Multi-Wavelength Graph

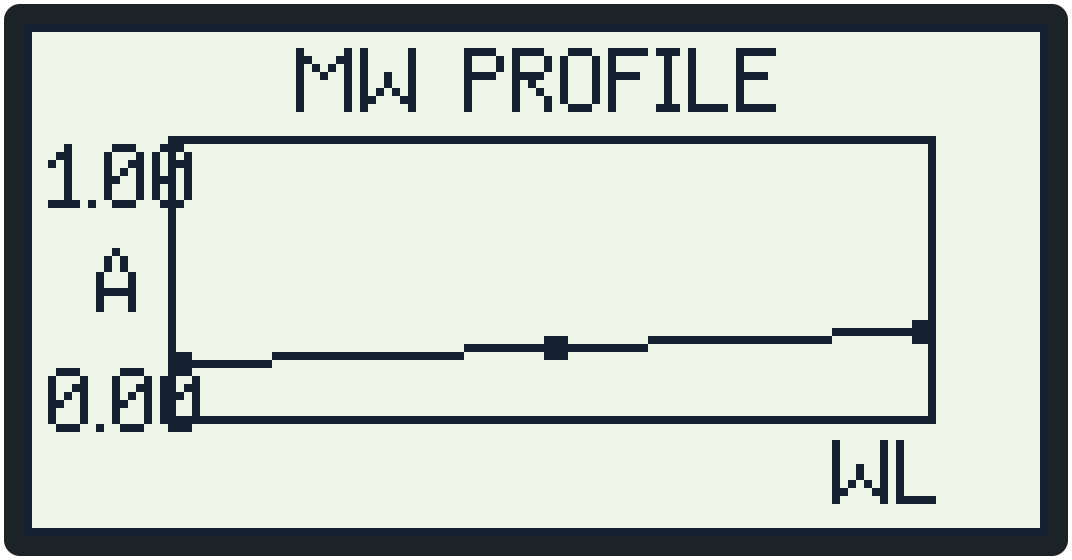
ID
multiWaveGraph

LEGACY WND
WND-29

MODE
Multi-wavelength analysis

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Multi-Wavelength Graph is used as a graph in the "Multi-wavelength analysis" group.

Information meaning: The window captures the "Multi-wavelength analysis" scenario state and links runtime id "multiWaveGraph" to unified index MW-GRAPH.

Content: LCD content is assembled from fields: multiWaveGraphData, selectedSignal.

Relation to other windows: Outgoing routes lead to: multiWaveJournal, saveDialog.

Source Screen IDs: multi-wave-graph

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`multiWaveGraphData`	array of graph points	Used by LCD renderer or handlers for screen "multiWaveGraph".
`selectedSignal`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "multiWaveGraph".

Transition routes

Action/button	Guard	Operation	Target screen
esc	—	runtime return to journal	multiWaveJournal · Multi-Wavelength Journal
file	—	runtime save graph	saveDialog · Save File

SET-MAIN · Settings

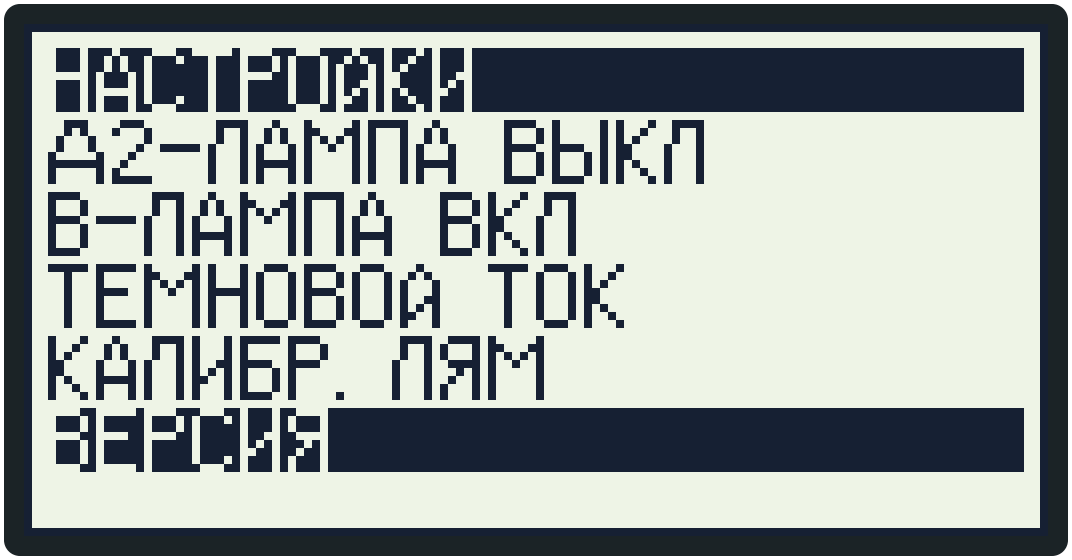
ID
settings

LEGACY WND
WND-30

MODE
Settings

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Settings is used as a menu in the "Settings" group.

Information meaning: The window captures the "Settings" scenario state and links runtime id "settings" to unified index SET-MAIN.

Content: LCD content is assembled from fields: settingsMenu, settingsIndex, d2Lamp, wLamp, slip, sampler.

Relation to other windows: Outgoing routes lead to: version, warning, main.

Source Screen IDs: 1-4 1-5

Source integrity: ambiguous

State table

Field	Type/meaning	Source/usage
`settingsMenu`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "settings".
`settingsIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "settings".
`d2Lamp`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "settings".
`wLamp`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "settings".
`slip`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "settings".
`sampler`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "settings".

Transition routes

Action/button	Guard	Operation	Target screen
enter	settingsIndex == 4	runtime version screen	version · Instrument Version
dark/wl	dark current or wavelength calibration result	runtime operation warning	warning · Warning
esc	—	runtime return	main · Main Menu

SET-VERSION · Instrument Version

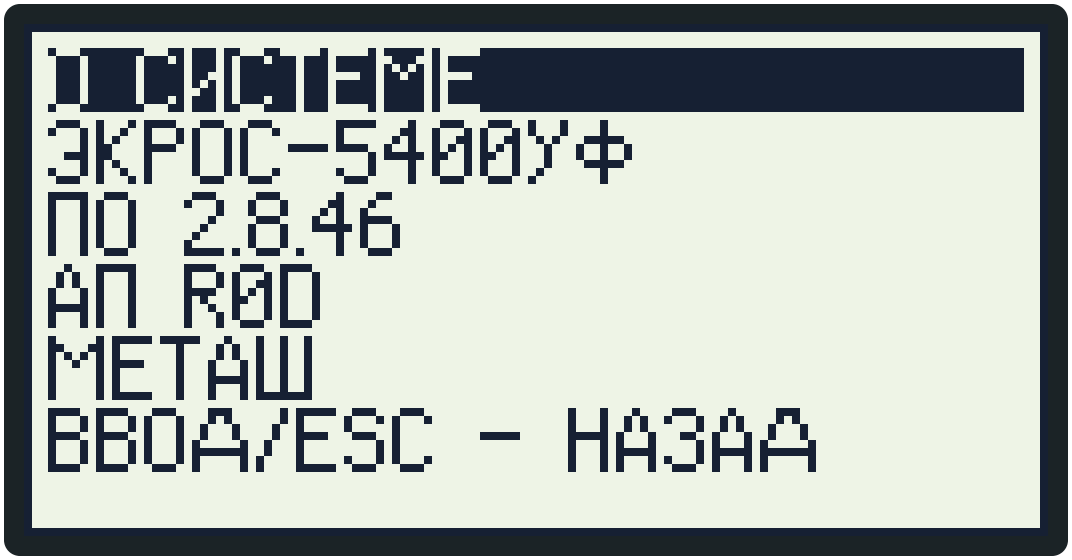
ID
version

LEGACY WND
WND-31

MODE
Settings

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Instrument Version is used as a system in the "Settings" group.

Information meaning: The window captures the "Settings" scenario state and links runtime id "version" to unified index SET-VERSION.

Content: LCD content is assembled from fields: softwareVersion, hardwareVersion, company.

Relation to other windows: Outgoing routes lead to: settings.

Source Screen IDs: 1-4 1-5

Source integrity: ambiguous

State table

Field	Type/meaning	Source/usage
`softwareVersion`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "version".
`hardwareVersion`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "version".
`company`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "version".

Transition routes

Action/button	Guard	Operation	Target screen
enter/esc	—	runtime return to settings	settings · Settings

FILE-ROOT · File Groups

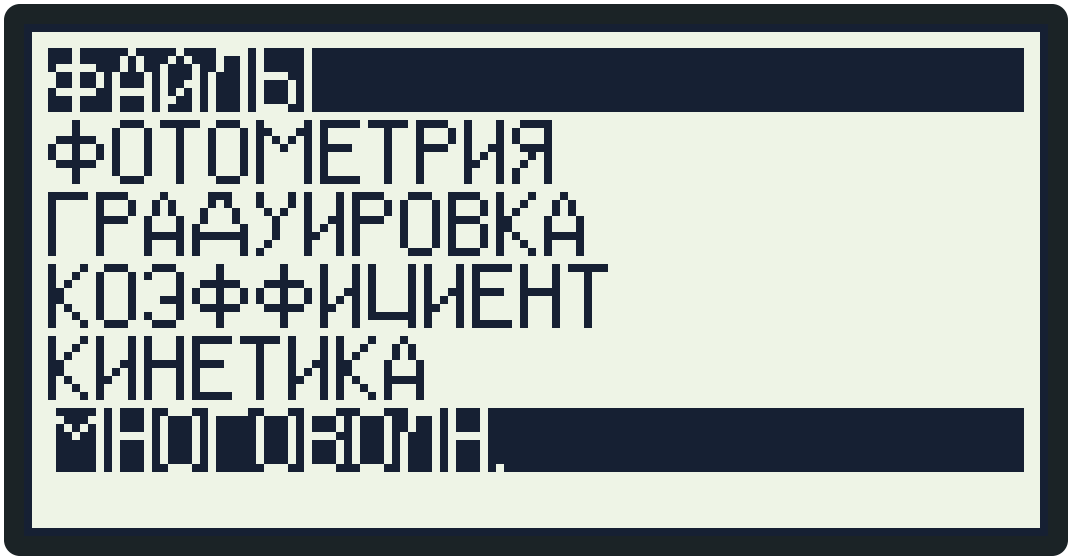
ID
fileRoot

LEGACY WND
WND-08

MODE
Files

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: File Groups is used as a file in the "Files" group.

Information meaning: The window captures the "Files" scenario state and links runtime id "fileRoot" to unified index FILE-ROOT.

Content: LCD content is assembled from fields: fileGroups, fileRootIndex.

Relation to other windows: Outgoing routes lead to: fileList, main.

Source Screen IDs: file-root

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`fileGroups`	file-manager context	Used by LCD renderer or handlers for screen "fileRoot".
`fileRootIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "fileRoot".

Transition routes

Action/button	Guard	Operation	Target screen
enter	—	runtime open group	fileList · File List
esc	—	runtime escape from root	main · Main Menu

FILE-LIST · File List

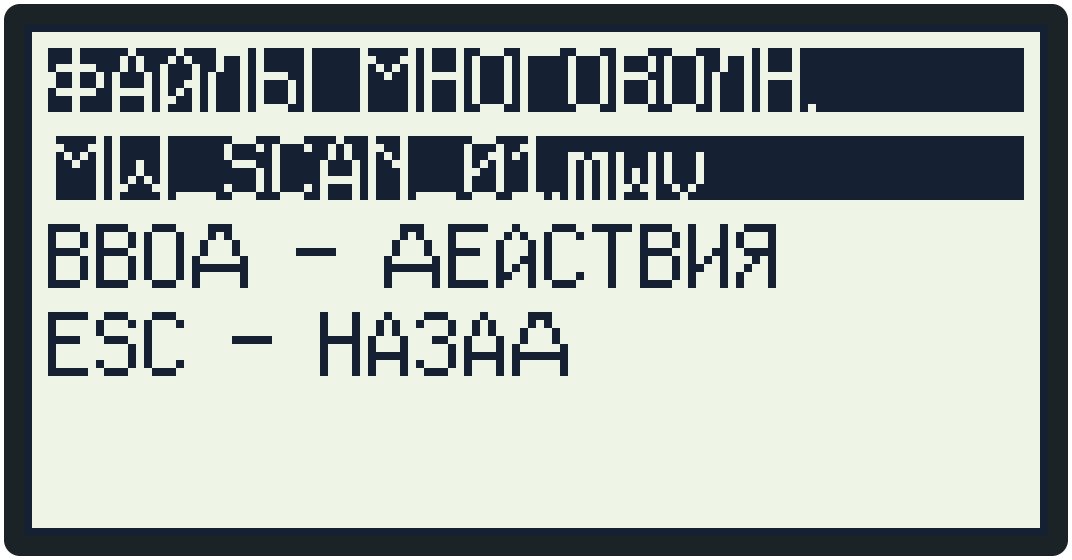
ID
fileList

LEGACY WND
WND-09

MODE
Files

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: File List is used as a file in the "Files" group.

Information meaning: The window captures the "Files" scenario state and links runtime id "fileList" to unified index FILE-LIST.

Content: LCD content is assembled from fields: fileContext, files, fileListIndex.

Relation to other windows: Outgoing routes lead to: fileActionMenu, fileRoot.

Source Screen IDs: file-list

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`fileContext`	file-manager context	Used by LCD renderer or handlers for screen "fileList".
`files`	file-manager context	Used by LCD renderer or handlers for screen "fileList".
`fileListIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "fileList".

Transition routes

Action/button	Guard	Operation	Target screen
enter	files exist	runtime file actions	fileActionMenu · File Actions
esc	previousScreen == main	runtime escape to root	fileRoot · File Groups

FILE-ACTION · File Actions

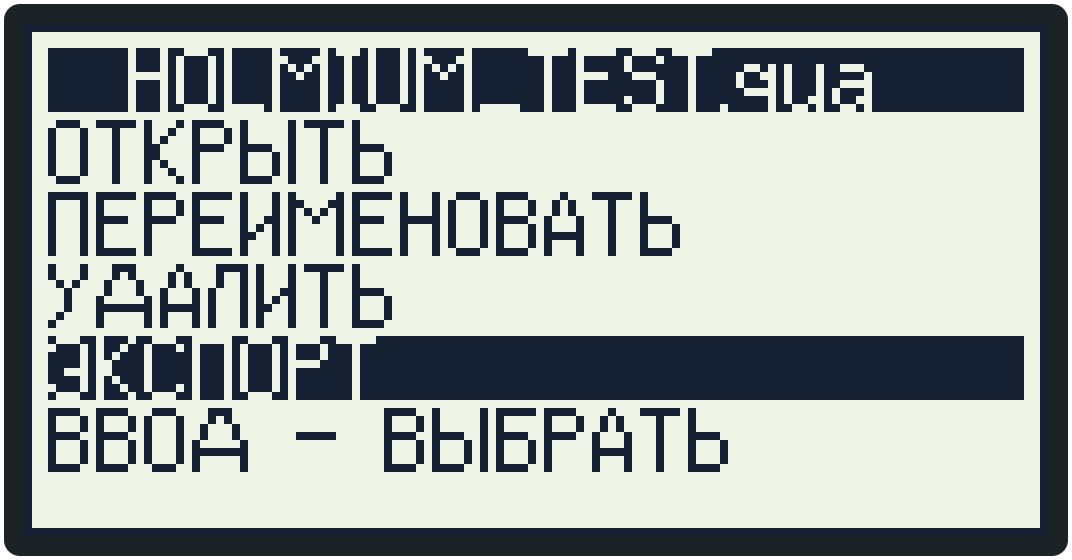
ID
fileActionMenu

LEGACY WND
WND-10

MODE
Files

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: File Actions is used as a file in the "Files" group.

Information meaning: The window captures the "Files" scenario state and links runtime id "fileActionMenu" to unified index FILE-ACTION.

Content: LCD content is assembled from fields: selectedFile, fileActions, fileActionIndex.

Relation to other windows: Outgoing routes lead to: input, warning, fileList.

Source Screen IDs: file-action

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`selectedFile`	file-manager context	Used by LCD renderer or handlers for screen "fileActionMenu".
`fileActions`	file-manager context	Used by LCD renderer or handlers for screen "fileActionMenu".
`fileActionIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "fileActionMenu".

Transition routes

Action/button	Guard	Operation	Target screen
rename	—	runtime rename dialog	input · Data Input
open/export	—	runtime file warning	warning · Warning
esc	—	runtime return to file list	fileList · File List

FILE-SAVE · Save File

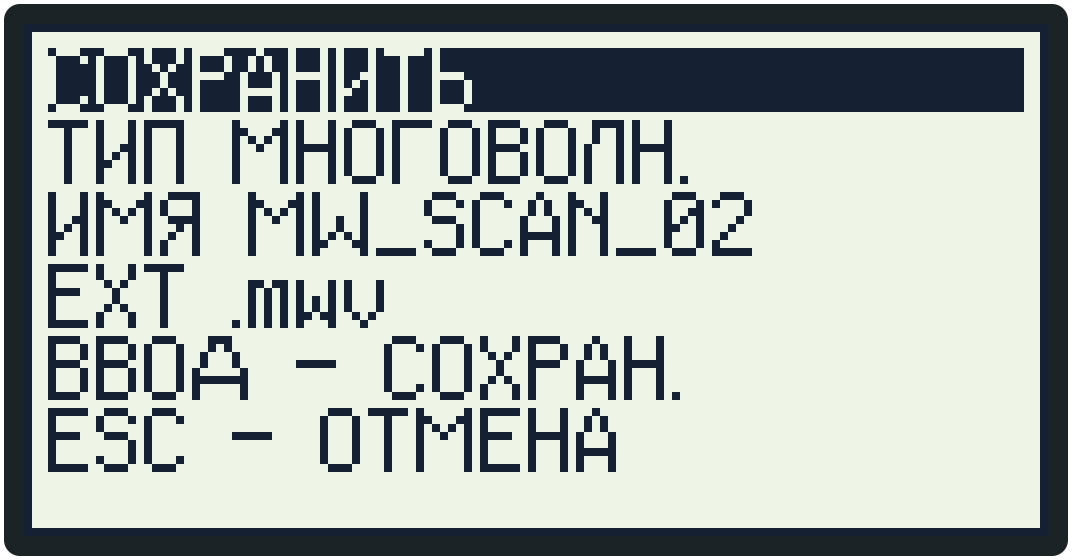
ID
saveDialog

LEGACY WND
WND-11

MODE
Files

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Save File is used as a dialog in the "Files" group.

Information meaning: The window captures the "Files" scenario state and links runtime id "saveDialog" to unified index FILE-SAVE.

Content: LCD content is assembled from fields: saveMeta, inputBuffer.

Relation to other windows: Outgoing routes lead to: main.

Source Screen IDs: save-dialog

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`saveMeta`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "saveDialog".
`inputBuffer`	string input buffer	Used by LCD renderer or handlers for screen "saveDialog".

Transition routes

Action/button	Guard	Operation	Target screen
esc/enter	actual target is returnScreen/previousScreen	runtime save result	main · Main Menu

MAIN-MENU · Main Menu

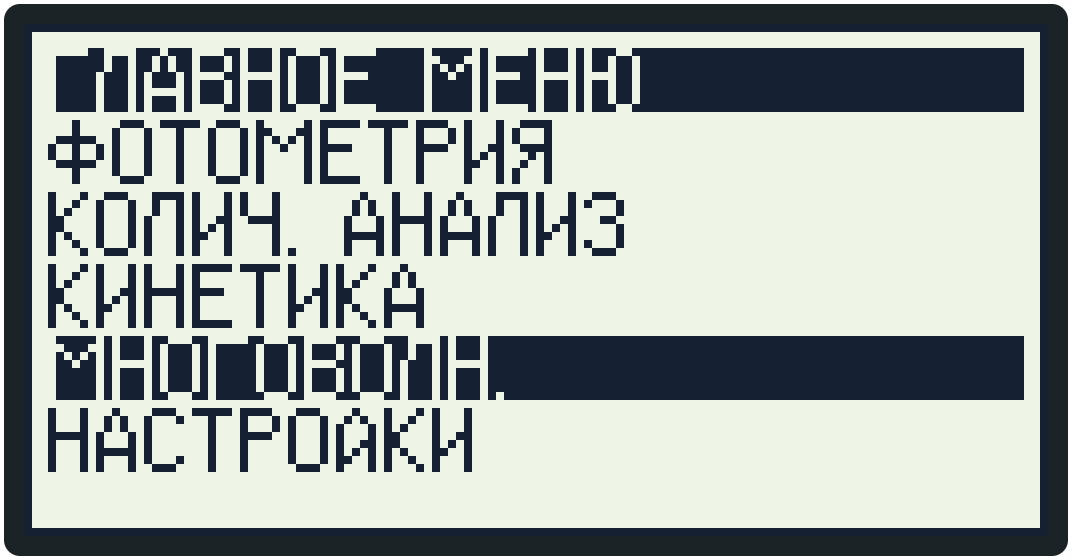
ID
main

LEGACY WND
WND-04

MODE
Shared screens

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Main Menu is used as a menu in the "Shared screens" group.

Information meaning: The window captures the "Shared screens" scenario state and links runtime id "main" to unified index MAIN-MENU.

Content: LCD content is assembled from fields: menuItems, mainIndex, wavelength, sampleLabel.

Relation to other windows: Outgoing routes lead to: photometry, quantMain, kineticsMenu, multiWaveMenu, settings, fileRoot.

Source Screen IDs: 1-4

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`menuItems`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "main".
`mainIndex`	numeric selected-item index	Used by LCD renderer or handlers for screen "main".
`wavelength`	number, wavelength in nm	Used by LCD renderer or handlers for screen "main".
`sampleLabel`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "main".

Transition routes

Action/button	Guard	Operation	Target screen
enter	mainIndex == 0	Screen 1-4 select Фотометрия	photometry · Photometry
enter	mainIndex == 1	Screen 1-4 select Колич. анализ	quantMain · Quantitative Analysis
enter	mainIndex == 2	Screen 1-4 select Кинетика	kineticsMenu · Kinetics Parameters
enter	mainIndex == 3	runtime multiwave menu	multiWaveMenu · Multi-Wavelength Analysis
enter	mainIndex == 4	Screen 1-4 select Настройки	settings · Settings
file	—	runtime file key	fileRoot · File Groups

SHARED-INPUT · Data Input

ID
input

LEGACY WND
WND-12

MODE
Shared screens

STATUS
implemented

LCD mockup

```
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Data Input is used as a input in the "Shared screens" group.

Information meaning: The window captures the "Shared screens" scenario state and links runtime id "input" to unified index SHARED-INPUT.

Content: LCD content is assembled from fields: dialogTitle, inputTarget, inputBuffer, returnScreen.

Relation to other windows: Outgoing routes lead to: main.

Source Screen IDs: 1-5-1-1-1 1-5-1-1-2 1-5-1-1-3

Source integrity: merged

State table

Field	Type/meaning	Source/usage
`dialogTitle`	state value or LCD placeholder	Used by LCD renderer or handlers for screen "input".
`inputTarget`	string input buffer	Used by LCD renderer or handlers for screen "input".
`inputBuffer`	string input buffer	Used by LCD renderer or handlers for screen "input".
`returnScreen`	contextual return screen	Used by LCD renderer or handlers for screen "input".

Transition routes

Action/button	Guard	Operation	Target screen
esc/enter	actual target is returnScreen	runtime input result	main · Main Menu

SHARED-WARNING · Warning

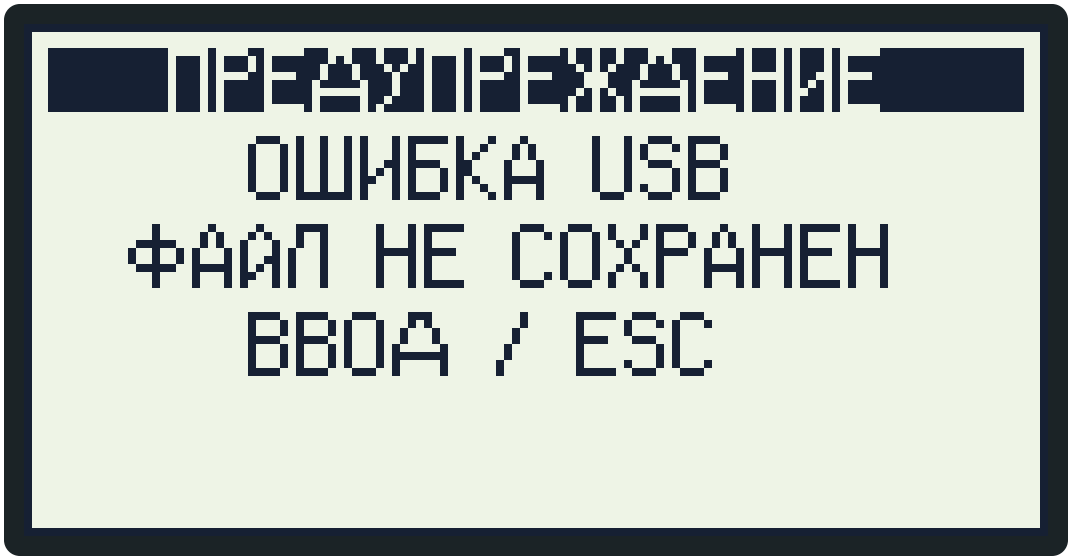
ID
warning

LEGACY WND
WND-32

MODE
Shared screens

STATUS
implemented

LCD mockup




```
[object Object]
[object Object]
[object Object]
[object Object]
```

Window description

Purpose: Warning is used as a dialog in the "Shared screens" group.

Information meaning: The window captures the "Shared screens" scenario state and links runtime id "warning" to unified index SHARED-WARNING.

Content: LCD content is assembled from fields: warning.title, warning.body, warningReturn.

Relation to other windows: Outgoing routes lead to: main.

Source Screen IDs: warning

Source integrity: runtime-only

State table

Field	Type/meaning	Source/usage
`warning.title`	warning structure	Used by LCD renderer or handlers for screen "warning".
`warning.body`	warning structure	Used by LCD renderer or handlers for screen "warning".
`warningReturn`	contextual return screen	Used by LCD renderer or handlers for screen "warning".

Transition routes

Action/button	Guard	Operation	Target screen
enter/esc	actual target is warningReturn	runtime warning return	main · Main Menu

Source Index Conflict Matrix

Screen ID	canonical runtime ids
1-4	main, settings, version
1-6-1	quantMain, quantUnits, quantCoef
1-6-1-1-32	quantCoef, calibrationGraph
1-6-1-1-33	quantCoef, calibrationGraph
1-6-1-1-34	quantCoef, calibrationGraph
1-6-1-1-27	calibrationJournal, kineticsRun, kineticsGraph
1-5	settings, version